Wellness Grocery Inventory Analysis

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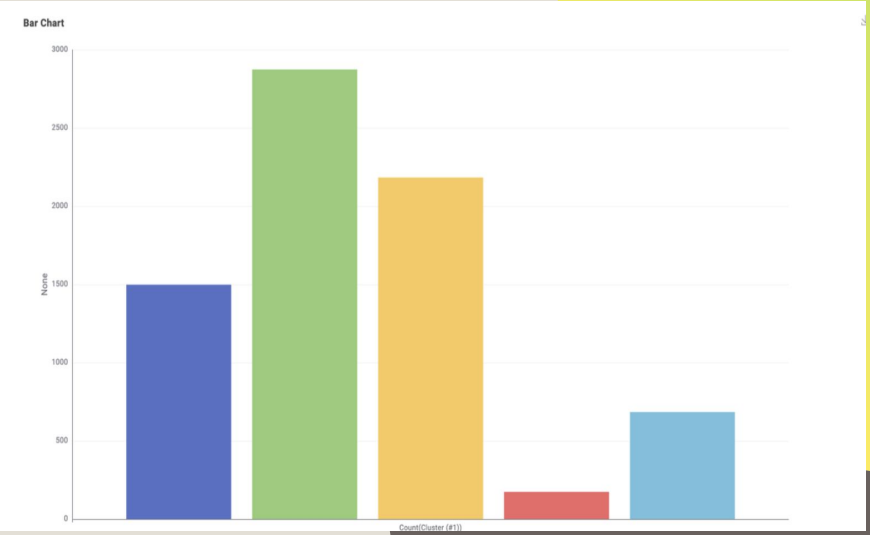
Our Objective

- Analyze inventory using data-driven methods
- Identify nutritional patterns across products
- Evaluate alignment with “wellness” expectations
- Use PCA and k-Means clustering

Dataset



Dataset includes products and features with 100+ products

Focused on nutritional attributes such as:

- Calories
- Sugar
- Protein
- Fat

Each row represents a product currently in inventory

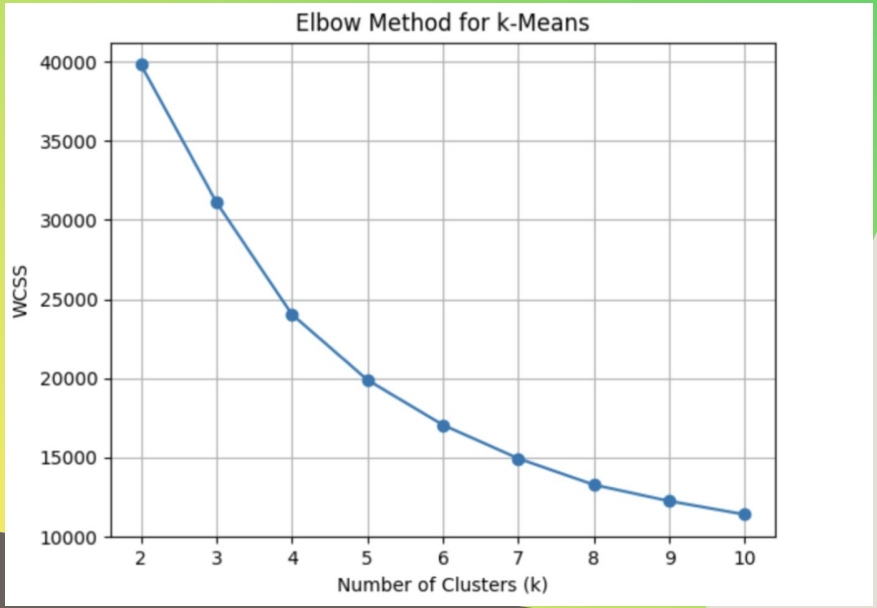
Methodology

- Data cleaning (missing values)
- Normalization (standard scaling)
- PCA (dimensionality reduction)
- k-Means clustering

PCA dimension 0 .00 Number (Float)	PCA dimension 1 .00 Number (Float)
5.7	5.808
5.656	5.75
7.39	7.057
1.408	2.523
1.522	2.603
1.137	2.44
0.789	2.166
1.547	2.635

Selecting k

- Tested k = 2 → 10
- Used WCSS to evaluate clusters
- Optimal k = 5

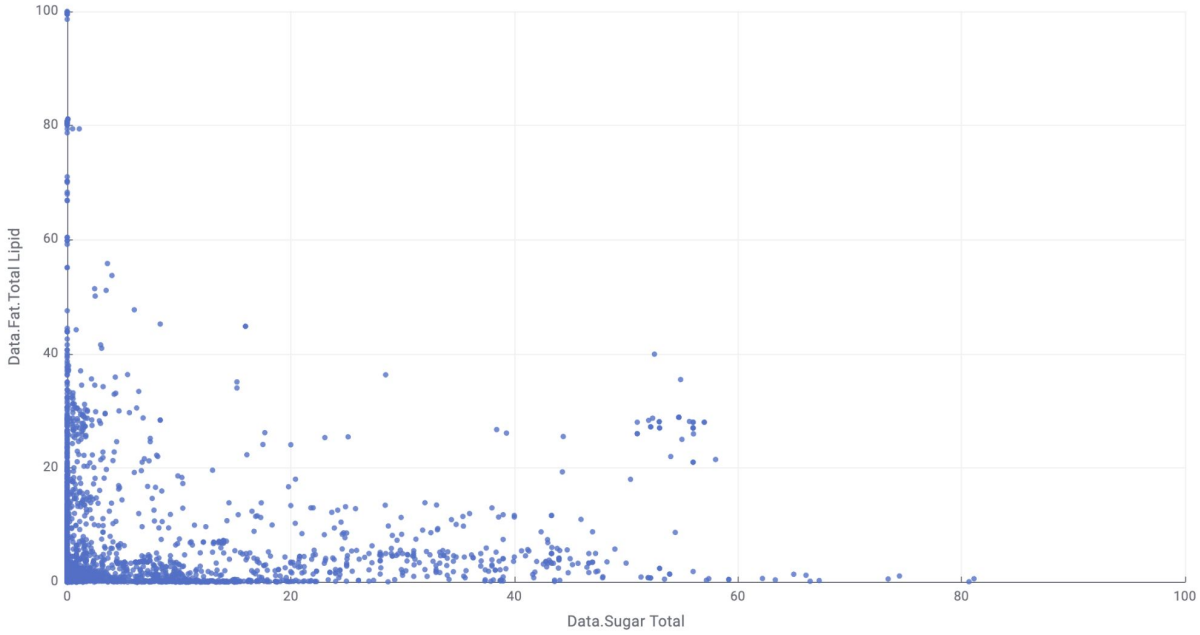


Outliers in our Inventory

Most products fall within moderate nutritional ranges

A small number of products are extremely high in sugar, fat, or calories

These items pull the distribution and distort clustering



Key Statistical Insights

Kilocalories: Mean ≈ 220, Max ≈ 902
Sugar: Mean ≈ 6.4, Max ≈ 99.9
Fat: Mean ≈ 10.2, Max ≈ 100
Fiber: Mean ≈ 2.0 (low)

Inventory Patterns

Our clustering analysis reveals 4 main product groups

- Sugary Snacks
  - High sugar and high calorie items
  - Includes candy, desserts, and processed snacks
- Healthy / Low-Calorie Items
  - Low sugar, lower calories
  - Typically align with wellness-focused products
- High-Protein Products
  - Moderate calories with higher protein
  - Includes protein bars, shakes, and supplements
- Mixed-Profile Items
  - Products that don’t clearly fit one category
  - Often combine high protein *and* high sugar

Cluster Validation

- Silhouette score ≈ 0.425
- Indicates moderate separation
- Some overlap expected

RowID	Mean Silhouette Coefficient .00 *Number (Float)*
cluster	0.382
cluster	0.149
cluster	0.546
cluster	0.437
cluster	0.306
Overall	0.425

Wellness Gaps

Where our inventory does not match our wellness goals:

- Some items labeled as “healthy” cluster with sugary snacks
- These products tend to have:
  - High sugar content
  - High calorie counts

Examples of concern:

- Protein bars with high sugar
- “Healthy” snacks that behave like desserts

What this means:

- Product labeling is not always accurate
- Customers may perceive items as healthy when they are not

Business Impact:

- Weakens brand trust
- Creates inconsistency in store positioning

Conclusion

Our data shows clear product groupings based on nutrition

Some products do not align with their intended category

Clustering helps identify:

- Misleading products
- Gaps in inventory

Any questions?
Ask away!

Thank you